CS-104 Quiz 1 (ANSWER KEY)

Fall 2026

Name: _____

This is an in-class, closed-book quiz (30 points). If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.

1. (5 points)

Give the value of each Python expression below (½ point each). Make the resulting type clear (e.g., `8` vs. `8.0` vs. `'8'`). If it raises an error instead, name the error (e.g., "IndexError").

```
>>> 'a' + 'b'
```
'ab' — example answer

```
>>> 17 // 5
```

```
>>> 17 % 5
```

```
>>> 2 ** 3 ** 2
```

```
>>> 4 * 2.0
```

```
>>> 'ab' * 3
```

```
>>> '5' + '5'
```

```
>>> int(3.9)
```

```
>>> 'python'[-2]
```

```
>>> len('python')
```

```
>>> str(4) + 4
```

Answers: `3`, `2`, `512`, `8.0`, `'abababab'`, `'55'`, `3`, `'o'`, `6`, `TypeError`.

2. (6 points)

For each code segment below, identify the error and briefly explain how to fix it (2 points each: 1 for the error, 1 for the fix).

a)

```
print('It's a wonderful world!')
```

b)

```
a = 1
b = -5
c = 6
x = 0
y = ax**2 + bx + c
```

c)

```
x = input("Enter a number")
x = x / 10
print(x)
```

a) The `'` in "It's" ends the string early (SyntaxError): use double quotes outside, or `\'`. b) `ax` and `bx` are read as variable names (NameError): write `a*x` and `b*x`. c) `input()` returns a string, so `/` fails (TypeError): use `float(input(...))`.

3. (4 points)

Consider the code below. What do the three `print` statements output?

```
a = [4, 8, 15]
a[0] = 16
a.append(23)
b = a[1] + a[2]
a[-1] = 42
print(a)
print(b)
print(len(a))
```

`[16, 8, 15, 42]`, then `23`, then `4`. Grading: 1 point per line, plus 1 point for all three correct.

4. (4 points)

Write one Python statement for each task below (1 point each).

Assume `inventory = {'apples': 12, 'pears': 5}`.

a) Add a new entry mapping `'plums'` to `8`.

b) Increase the number of apples by 3, using its current value (don't just write 15).

c) Remove the entry for `'pears'` completely.

d) Print how many entries `inventory` has.

a) `inventory['plums'] = 8` **b)** `inventory['apples'] = inventory['apples'] + 3` **c)** `del inventory['pears']` **d)** `print(len(inventory))`

5. (6 points)

Write a code segment that inputs a number of total seconds from the user (as an integer) and prints how many whole hours, remaining minutes, and remaining seconds that is. Here is an example run:

```
Total seconds: 3725
1 hours, 2 minutes, 5 seconds
```

Note that you should:

- implement the prompt as shown.
- use `//` and `%` as appropriate.
- print the result in the format shown above.

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

```
total = int(input("Total seconds: "))
hours = total // 3600
minutes = (total % 3600) // 60
seconds = total % 60
print(hours, "hours,", minutes, "minutes,", seconds, "seconds")
```

Grading: input/int conversion 1, hours 1, minutes 2, seconds 1, output format 1.

6. (5 points)

Write a code segment that builds a username. It inputs the user's first name, last name, and birth year (keep the year as a string), and then prints a username made of:

- the first letter of the first name,
- followed by the whole last name,
- followed by the **last two** digits of the birth year (use indexing),

all in lowercase. Example run:

```
First name: Grace
Last name: Hopper
Birth year: 1906
Username: ghopper06
```

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

```
first = input("First name: ")
last = input("Last name: ")
year = input("Birth year: ")
username = (first[0] + last + year[-2] + year[-1]).lower()
print("Username:", username)
```

Grading: three inputs 1, `first[0]` 1, last name 1, `year[-2] + year[-1]` (or `year[2] + year[3]`) 1, lowercase 1.